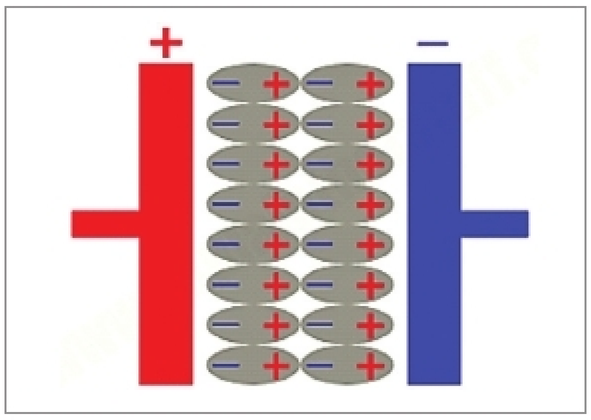

Fig. 1: Schematic diagram of ordinary capacitor

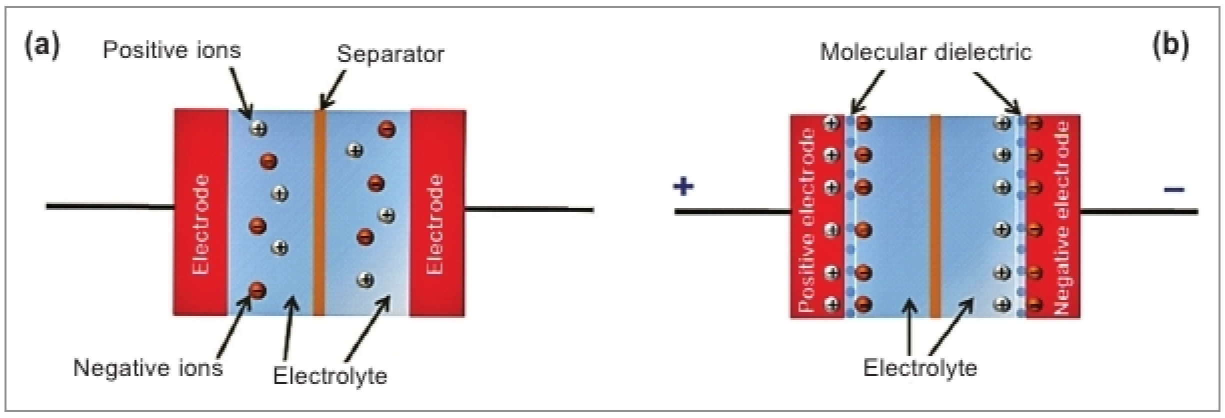

Fig. 2: Double layer supercapacitor: (a) discharged, (b) charged

between a conductive electrode and an adjacent liquid electrolyte. At the boundary, two layers of charge with opposing polarity are formed, one at the surface of the electrode and one in the electrolyte.

These two layers are typically separated by a single layer of solvent molecules (generally one molecule thick) that adhere to the surface of the electrode and act like a dielectric as in a conventional capacitor. The solvent molecules act as a barrier and prevent combination of opposite charges. Therefore no electric charge flows in between electrode and electrolyte. In conventional capacitor, the thickness of dielectric might range from a few microns to a millimetre or more.

In short, supercapacitors get their much bigger capacitance from a combination of plates with a bigger surface area and less distance between them (because of the very effective double layer). The supercapacitor's capacitance is in direct proportion to the area of electrical double layer.

Upon voltage application, the electrical double layer is created that aligns both negative and positive charges along the boundaries of electrodes and electrolytic solution. When these negative or positive ions migrate close to the electrode, they experience strong opposition from the solvent molecules. Hence, no charge transfer takes place between electrolyte and electrode.

However, these opposite charges exert electrostatic forces on each other. Thus, a large amount of charge is built up at the common boundary of electrode and electrolyte. The activated carbon layer deposited on the electrodes is very much porous and provides astounding surface area and acts as a storehouse for storing electric charges during charging.

Pseudo-capacitor. A pseudo-capacitor is a hybrid between a battery and electric double-layer capacitor. It also consists of two electrodes separated by an electrolyte. Charge storage occurs by chemical and electrostatic means. The chemical process involves charge transfer by means of reduction-oxidation that is redox reaction when most of the charge is transferred near the surface of the electrode.

On the other hand, in double-layer capacitors, positive and negative charges reside on two electrode surfaces. While the charge transfer is similar to that in a battery, transfer rates are higher because of the use of thinner redox material on the electrode, or lower penetration of the ions from the electrolyte into the structure.

Performance Parameters

Parameters	Electrolytic capacitors	Double layer supercapacitors	Pseudocapacitors and hybrid type (Li-ion)
Temperature range (°C)	-40 to +125	-20 to +70	-20 to +70
Maximum charge (V)	4 to 630	1.2 to 3.3	2.2 to 3.8
Recharge cycles (thousands k)	Unlimited	100k to 1000k	20k to 100k
Capacitance (Farads)	Up to 2.7	100 to 12,000	300 to 3300
Specific energy (Watt hours per kg)	0.01 to 0.3	4 to 9	10 to 15
Self-discharge time	Short, in days	In weeks	Long, in months
Efficiency (%)	99	95	90
Working life in years	20	5 to 10	5 to 10

Because of multiple processes acting to store charge, the capacitance values are higher. Pseudocapacitance is accompanied by an electron charge transfer between electrolyte and electrode coming from a de-solvated and adsorbed ion. The adsorbed ion has no chemical reaction with the atoms of the electrode since only a charge transfer takes place. It uses transition metal-oxides like RuO_2, IrO_2, or MnO_2 inserted by doping in the conductive material like active carbon, which covers the electrode. Both pseudocapacitance and double-layer capacitances contribute inseparably to the total capacitance value.

Hybrid supercapacitor. Hybrid supercapacitors are the devices with elevated energy storage capability and elevated capacitance. It combines electric double-layer capacitor and lithium-ion technology, resulting in an energy density up to 115 per cent higher. The hybrid capacitor offers an enhanced capability of energy storage, providing power to applications more quickly and efficiently.

Construction details

Electrodes. The internal arrangement of a wound type cylindrical supercapacitor is shown in Fig. 3. It comprises active layers made from activated carbon electrodes and separator with aluminium tabs inserted in-between active layers. Aluminium foils should be able to distribute a peak current of 100A, and act as the current collector. They are coated with activated carbon, which is electrochemically etched, so that the surface area of the material is about 100,000 times greater than the smooth surface. The electrodes are separated by a very thin ion-permeable separator made of porous polymeric films or woven ceramic fibres.